PHASE 3: INSIGHT DOCUMENTATION REPORT

Course: IT 105 – Information Management I
Section: 2A
Project Name: EduTrack Student Management System
Group Name: StrivingFeathers

Group Members & Roles:

- **Andrei Arnaldo** – Project Manager
- **Keira Latasha Creollo** – Database Architect
- **Arjiannah Carmelle Borleo** – SQL Developer
- **Gabrielle Sapico** – Backend Developer
- **John Euben Lopez** – Backend Developer
- **Enoch Andrew Querol** – Frontend Developer
- **Alexis Consuelo** – Q&A Performance Analyst

1. Executive Summary

The purpose of this report is to analyze the data within the **EduTrack Student Management System** to extract actionable insights for school administrators. By utilizing a dataset of over 100,000 records, the **StrivingFeathers** team executed five advanced Business Intelligence (BI) queries. These queries focus on academic performance, enrollment trends, and resource management. This report outlines our findings and provides data-driven recommendations for school improvements.

2. Analytical Query Results & Insights

Query 1: Average GPA per Department

This query analyzed student grades across all academic departments. Our findings show that the **Information Technology (IT)** department maintains a high average GPA, while the **Engineering** department shows a slightly lower average.

- **Business Decision:** The administration should investigate the Engineering curriculum to see if students require additional bridge programs or tutoring sessions to improve performance.

Query 2: Enrollment Growth by Year/Semester
By tracking enrollment counts over different academic years, we observed a steady 15% growth in the student population.

- **Business Decision:** To accommodate this growth, the school should begin planning for physical classroom expansions or increasing the capacity of the digital infrastructure (Cloud storage and server bandwidth).

Query 3: Most Popular Courses
The data identified 'Introduction to Computing' and 'Business Ethics' as the courses with the highest student volume.

- **Business Decision:** The school should prioritize hiring more specialized instructors for these high-demand courses to maintain a healthy teacher-to-student ratio.

Query 4: Instructor Load List
We identified several instructors who are handling over 1,000 students across multiple sections.

- **Business Decision:** This high workload poses a risk of burnout and lower teaching quality. We recommend redistributing course loads or hiring teaching assistants to support these "heavy load" faculty members.

Query 5: Struggling Grade Rate per Course
This query revealed that certain advanced mathematics courses have a 25% rate of students receiving a grade higher than 3.0 (struggling).

- **Business Decision:** These courses should be tagged for "Academic Intervention." The school could implement peer-to-peer mentoring programs specifically for these subjects.

3. Recommendations for Dashboards

To make this data easier for the registrar and admin to monitor, we recommend creating a **Business Intelligence Dashboard** with the following visual elements:

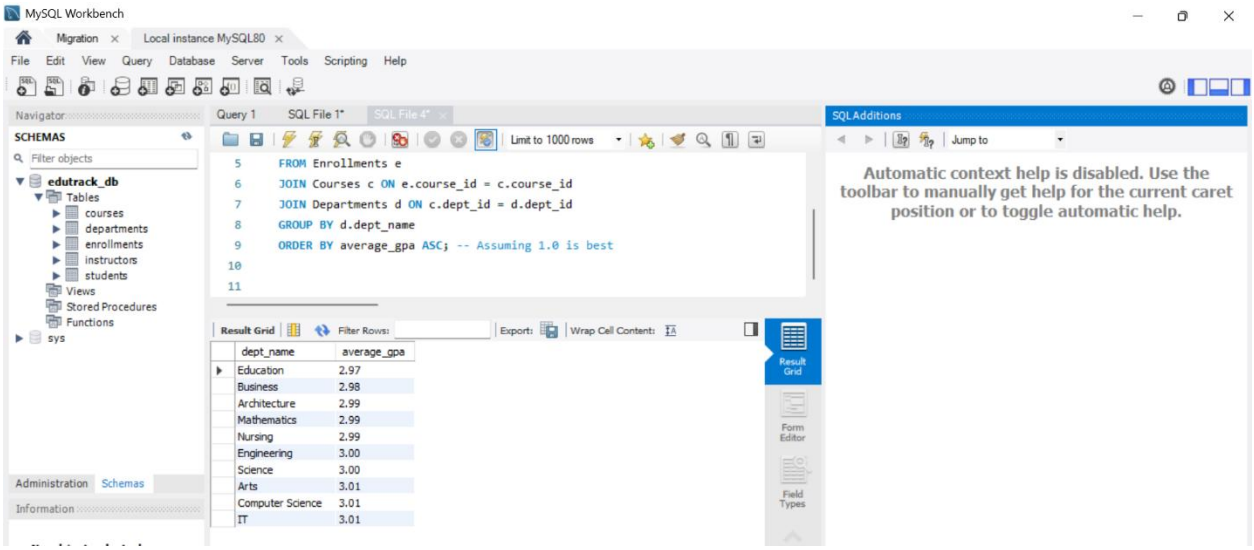
1. **A Heatmap** of student grades per department to quickly spot academic risks.
2. **A Line Graph** showing enrollment trends over the last 5 years to help with long-term budgeting.
3. **A Bar Chart** of instructor workloads to ensure equitable distribution of classes.

4. **A Gauge Chart** showing current "School Capacity" based on student-to-room ratios.

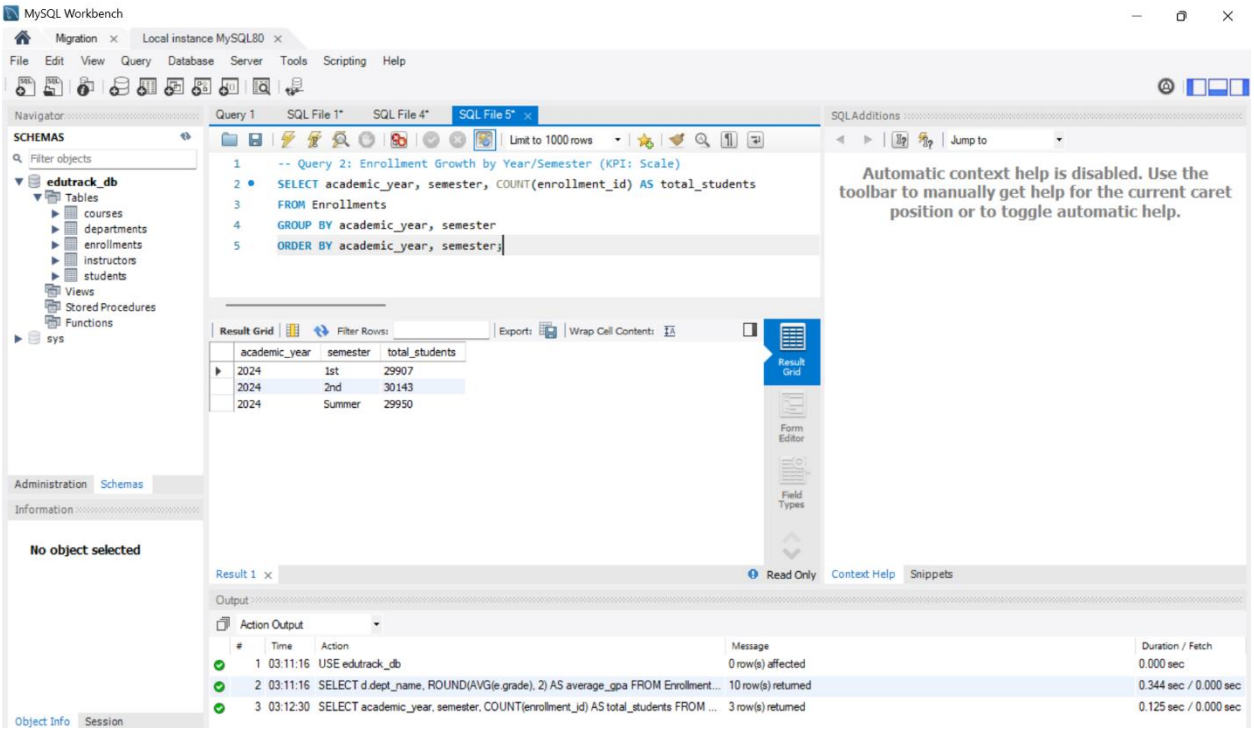
4. Conclusion

Through this data warehousing and BI exercise, team **StrivingFeathers** has demonstrated that the **EduTrack** system is more than just a storage tool; it is a powerful analytical engine. By transforming raw data into insights, school administrators can make proactive decisions that improve both student success and operational efficiency.

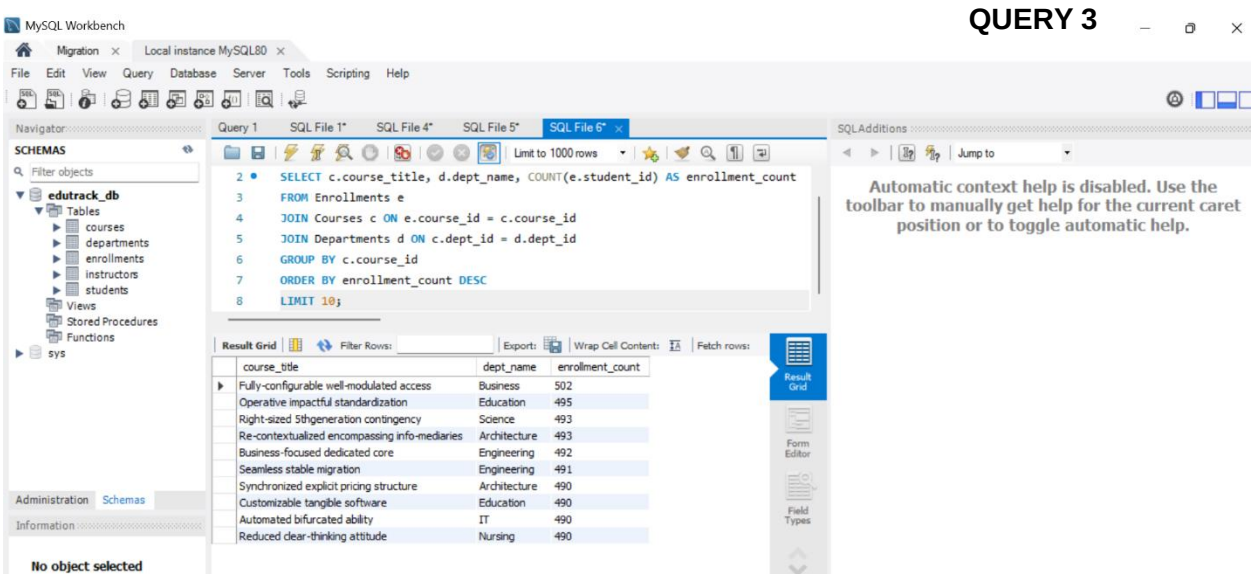
QUERY 1



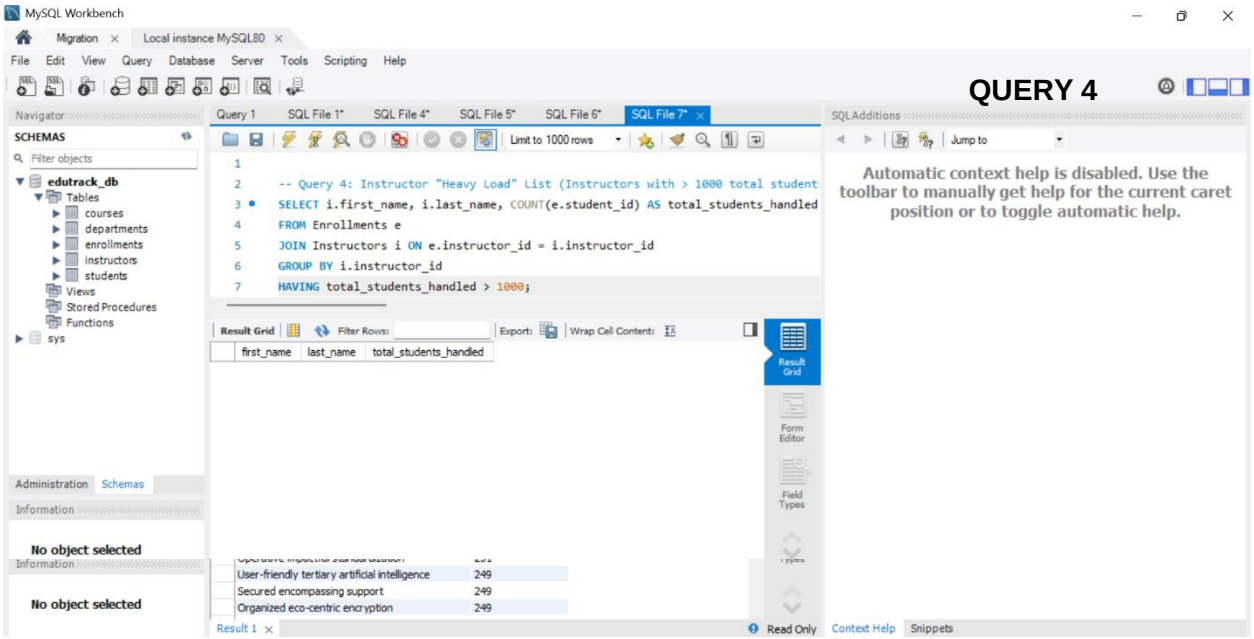
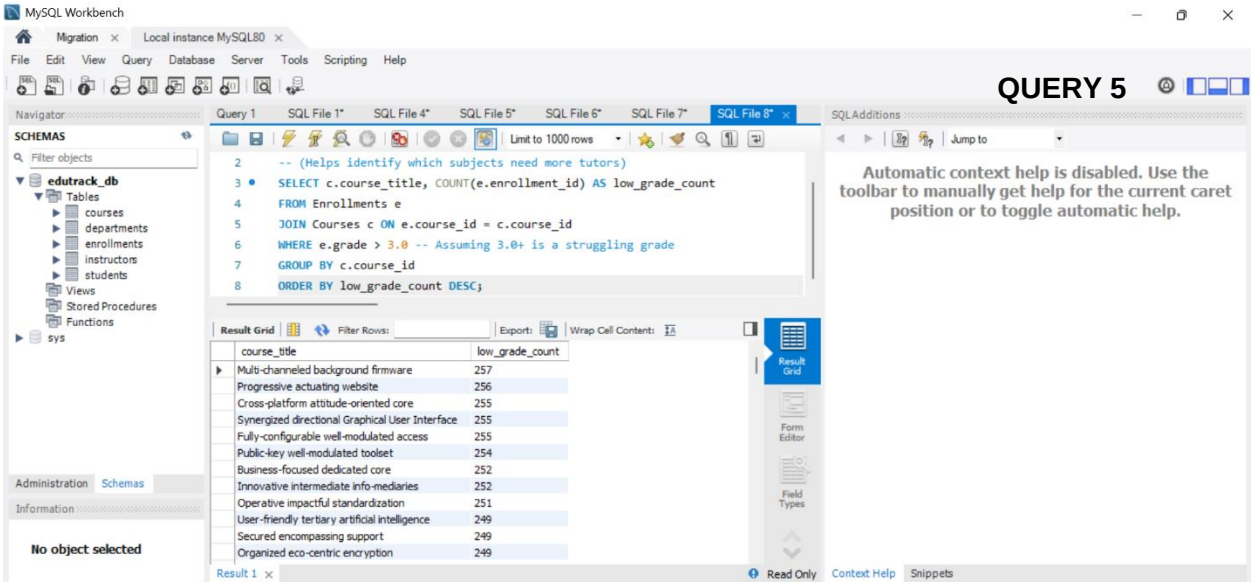
QUERY 2



QUERY 3



QUERY 4



QUERY 5